Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following is not a correct statement?

(A) Ionic compounds are electrically neutral.

(B) Boiling point of an ionic compound is more than a covalent compound.

(C) Melting point of a covalent compound is more than an ionic compound.

(D) Ionic compounds are soluble in polar solvent.

Q.2

In PO_4^{3-}, the formal charge on each oxygen atom and the P - O bond order respectively are

(A) $-0.75, 0.6$

(B) $-0.75, 1.0$

(C) $-0.75, 1.25$

(D) $-3, 1.25$

Q.3

Which of the following statements is incorrect?

(A) NH_3 is more basic than PH_3.

(B) NH_3 has a higher boiling point than that of HF.

(C) N_2 is less reactive than P_4.

(D) The dipole moment of NH_3 is less than that of SO_2

Q.4

CaO and $NaCl$ have the same crystal structure and approximately the same ionic radii. If U is the lattice energy of $NaCl$, the approximate lattice energy of CaO is

(A) $U/2$

(B) U

(C) $4U$

(D) $2U$

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Q.5

The compound which contains both ionic and covalent bonds is

(A) CH_4

(B) H_2

(C) KCN

(D) KCl

Q.6

The electronegativity difference between N and F is greater than that between N and H yet the dipole moment of $NH_3(1.5D)$ is larger than that of $NF_3(0.2D)$. This is because

(A) in NH_3 the atomic dipole and bond dipole are in the same direction whereas in NF_3 these are in opposite directions.

(B) in NH_3 as well as NF_3 the atomic dipole and bond dipole are in opposite directions.

(C) in NH_3 the atomic dipole and bond dipole are in the opposite directions whereas in NF_3 these are in the same direction.

(D) in NH_3 as well as in NF_3 the atomic dipole and bond dipole are in the same direction.

Q.7

The dipole moments of diatomic molecules AB and CD are $10.41D$ and $10.27D$, respectively while their bond distances are 2.82 and $2.67\mathring{A}$, respectively. This indicates that

(A) bonding is 100% ionic in both the molecules

(B) AB has more ionic bond character than CD

(C) AB has lesser ionic bond character than CD

(D) bonding is nearly covalent in both the molecules

Q.8

Bond distance in HF is 9.17×10^{-11} m. Dipole moment of HF is $6.104 \times 10^{-30}Cm$. The percentage ionic character in HF will be (electron charge $= 1.60 \times 10^{-19}C$)

(A) 61.0%

(B) 38.0%

(C) 35.5%

(D) 41.5%

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Q.9

Dipole moment is shown by

(A) 1,4-dichlorobenzene

(B) cis $-1,2$ -dichlorobenzene

(C) trans -1, 3-dichlorobenzene

(D) trans $-2,3$ -dichloro -2 - butene

Q.10

What is the $\%$ ionic character of HCl if its observed dipole moment is $1.03 \mathrm{D}$, electronic charge (q) is 4.8×10^{-10} e.s.u. and distance (d) between atoms is $1.27 \mathring{A}$?

(A) 63% ionic

(B) 17% ionic

(C) 83% ionic

(D) 89% ionic

Q.11

Which of the following statements is correct in the context of the allene molecule, $\mathrm{C_3H_4}$?

(A) The central carbon is sp hybridized

(B) The terminal carbon atoms are sp^2 hybridized

(C) The planes containing the $\mathrm{CH_2}$ groups are mutually perpendicular to permit the formations two separate π -bonds

(D) All are correct

Q.12

The formation of molecular complex $\mathrm{BF_3-NH_3}$ results in a change in hybridization of boron

(A) from sp^2 to dsp^2

(B) from sp^2 to sp^3

(C) from sp^3 to sp^2

(D) from sp^3 to sp^3d

Q.13

The compound showing maximum covalent character is

(A) $\mathrm{BF_3}$

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(B) BBr_3

(C) BCl_3

(D) BI_3

Q.14

Amongst $LiCl, RbCl, BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are:

(A) $LiCl$ and $RbCl$

(B) $RbCl$ and $BeCl_2$

(C) $MgCl_2$ and $BeCl_2$

(D) $RbCl$ and $MgCl_2$

Q.15

Which one of the following molecules is polar?

(A) XeF_4

(B) IF_5

(C) SbF_5

(D) CF_4

Answers & Solutions

Q.1 Answer:

Melting point of a covalent compound is more than an ionic compound.

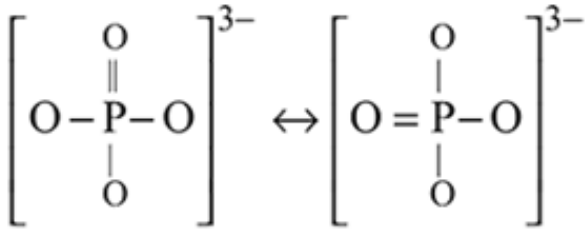
Solution:

Due to electrostatic force of attraction, the melting points and boiling points of ionic compounds are high.

Q.2 Answer:

$-0.75, 1.25$

Solution:



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Bond order
$= \frac{\text{Number of bonds}}{\text{Number of Resonating structures}}$
$= \frac{5}{4} = 1.25$
Three unit negative charge is being shared by four O atoms. Formal charge $= -3/4 = -0.75$

Q.3 Answer:

NH_3 has a higher boiling point than that of HF.

Solution:

Due to high EN of F than N, H -bonding is stronger in HF than NH_3 thus b.pt. of HF $> NH_3$.

Q.4 Answer:

$4U$

Solution:

Lattice energy $= \frac{\text{Product of charges}}{\text{interionic distance}}$
In $NaCl$ the product of charges $= 1 \times 1$; In CaO product of charges $= 2 \times 2 = 4$ while the inter ionic distance is almost same in both. Thus lattice energy of CaO is almost four times the lattice energy of $NaCl$.

Q.5 Answer:

KCN

Solution:

KCN molecule contains both ionic and covalent bonds, as follows
$K^+[C \equiv N]^-$
$\uparrow$
covalent bond

Q.6 Answer:

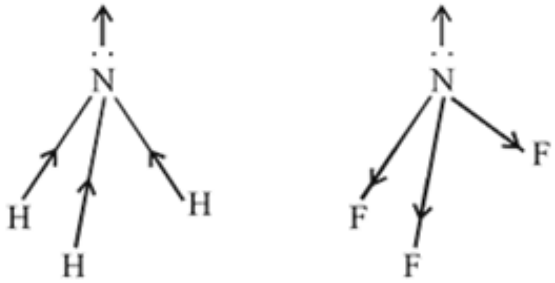
in NH_3 the atomic dipole and bond dipole are in the same direction whereas in NF_3 these are in opposite directions.

Solution:

In NH_3 the atomic dipole (orbital dipole due to lone pair) and bond dipole are in the same direction whereas in NF_3 these are in opposite direction so in the former case they are added up whereas in

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the latter case net result is reduction of dipole moment. It has been shown in the following figure :



Q.7 Answer:

AB has lesser ionic bond character than CD

Solution:

As dipole moment = electric charge × bond length

D. M. of AB molecule

$= 4.8 \times 10^{-10} \times 2.82 \times 10^{-8} = 13.53\mathrm{D}$

D.M. of CD molecule

$= 4.8 \times 10^{-10} \times 2.67 \times 10^{-8} = 12.81\mathrm{D}$

Now \% ionic character

$= \frac{\text{Actual dipole moment of the bond}}{\text{Dipole moment of pure ionic compound}} \times 100$ **then % ionic character in** $\mathrm{AB} = \frac{10.41}{13.53} \times 100 = 76.94\%$

% ionic character in CD

$= \frac{10.27}{12.81} \times 100 = 80.17\%$

Q.8 Answer:

41.5%

Solution:

Given $e = 1.60 \times 10^{-19}\mathrm{C}$

$d = 9.17 \times 10^{-11}\ \mathrm{m}$

From $\mu = e \times d$

$\mu = 1.60 \times 10^{-19} \times 9.17 \times 10^{-1}$

$= 14.672 \times 10^{-30}$

\% ionic character

$= \frac{\text{Observed dipole moment}}{\text{Calculated Dinole moment}} \times 100$

$= \frac{6.104 \times 10^{-30}}{14.672 \times 10^{-30}} \times 100$

$= 41.5\%$

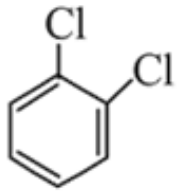
Q.9 Answer:

cis $-1,2$ -dichlorobenzene

Solution:

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Cis 1,2 - dichloro benzene will have some

D.M. 

Q.10 Answer:

17% ionic

Solution:

$$\mu_{\text{ionic}} = \text{q} \times \text{d}$$
$$= 4.8 \times 10^{-10}\text{esu} \times 1.27 \times 10^{-8}\ \text{cm}$$
$$\Rightarrow 6.096 \times 10^{-18}\text{esucm}$$

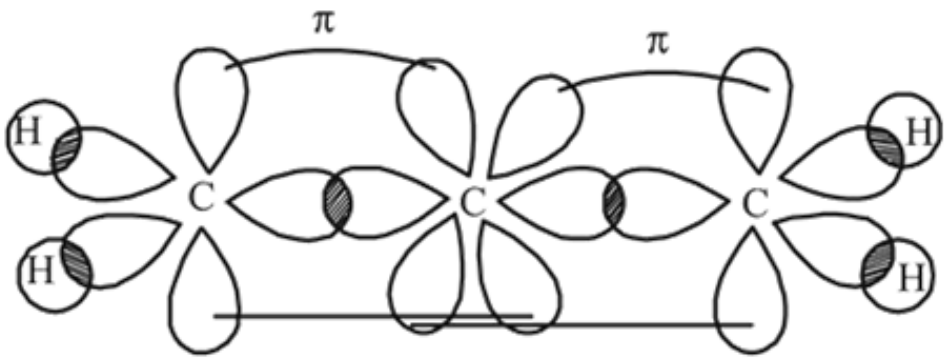
\% Ionic character

$$= \frac{\text{Actual dipole moment of the bond} \times 100}{\text{dipole moment of a pure ionic bond}}$$
$$\Rightarrow \frac{1.03 \times 10^{-18}}{6.096 \times 10^{-18}} \times 100 = 17\%$$

Q.11 Answer:

All are correct

Solution:



Q.12 Answer:

from sp^2 to sp^3

Solution:

In BF_3, B is sp^2 hybridized with one empty p_z orbital. The empty p_z orbital of BF_3 can be filled by lone pair of molecules such as NH_3. When this occurs a tetrahedral molecule or ion is formed which is sp^3 hybridized.

Q.13 Answer:

BF_3

Solution:

BF_3 show maximum covalent character due to back bonding.

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Q.14 Answer:

$RbCl$ and $BeCl_2$

Solution:

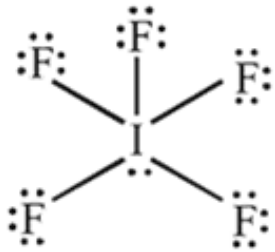
According to Fajan's rules smaller, highly charged cation has greatest covalent character while large cation with smaller charge has greatest ionic character.

Q.15 Answer:

IF_5

Solution:

The geometry of IF_5 is square Pyramidal with an unsymmetric charge distribution therefore this molecule is polar.



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